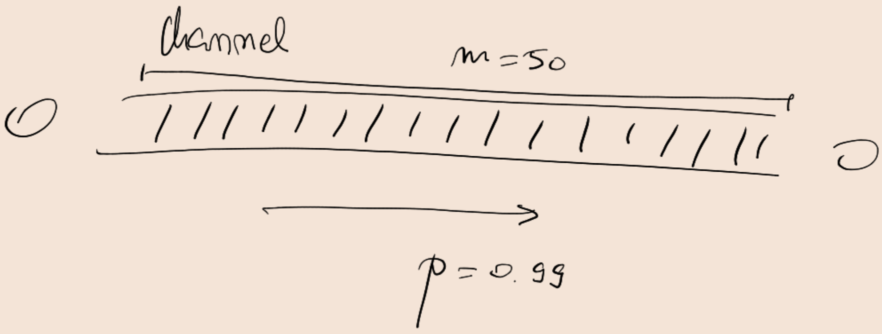


$$C_n^t \cdot p^{n-t} (1-p)^t$$

$t=0$: Prob. of 0 errors: $C_{50}^0 \cdot p^{50-0} \cdot \underbrace{(1-0.99)^0}_{1} = 1 \cdot p^{50} \cdot 1 = p^{50} = (0.99)^{50} = 0.605$

$t=1$: Prob. of 1 error: $C_{50}^1 \cdot 0.99^{50-1} \cdot (1-0.99)^1 = 50 \cdot (0.99)^{49} \cdot 0{,}1 = 0.3056\ldots$

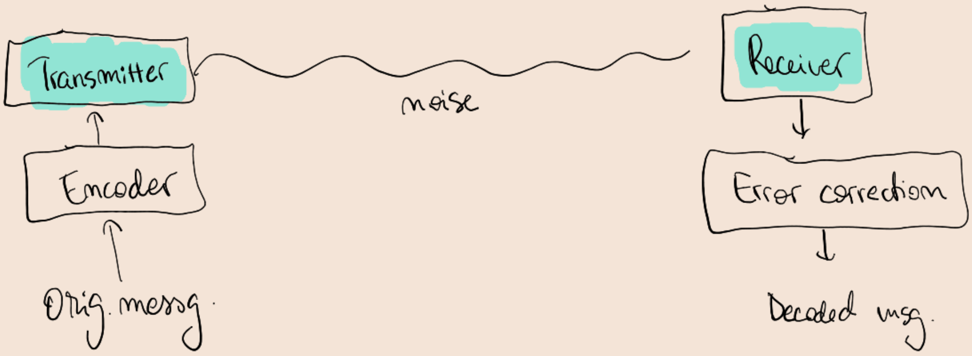
EAN-13 — ISBN (books), UPC (prod.)

$$\boxed{a_{13} = 10 - (a_1 + 3a_2 + a_3 + 3a_4 + \ldots + a_{11} + 3a_{12}) \% 10}$$

▌ → 1

▯ → 0

Error-correcting codes — CD, DVD, Blu-Ray; Credit cards; sattelite



Coding problem

↳ BINARY CODES → over finite fields.

SYMM. CHANNELS

$1 \xrightarrow{p_1} 0$

$0 \xrightarrow{p_2} 1$ $\quad p_1 = p_2$ (%)

| Mia | → | (n, 3) | → | n-3 check digits | 3 info digits. |

message of 3 digits — encoder — code word of n digits.

? 2^3 messages → 2^3 code words

$2 \cdot 2 \cdot 2$ — 2^n possible words.

Mia, ima, ami
Mai, iam, aim

balance k, $n-k$

Ex: (3,2) - parity check code

sum modulo 2 = XOR

ENCODE

msg = 00

Code word? → $(0+0)\%2 = 0\%2 = \boxed{000}$

msg = 01 → $(0+1)\%2 = 1\%2 = \boxed{101}$

msg = 10 → $(1+0)\%2 = 1\%2 = \boxed{110}$

msg = 11 → $(1+1)\%2 = 2\%2 = 011$

$\frac{1|2}{0|0}$
1

DECODE

↓ how are they received?

000 $\xrightarrow{\text{parity?}}$ no (we have no 1's)

101 $\xrightarrow{\text{parity?}}$ yes (we have an even number (2) of 1's)

110 → yes

011 → yes ← why 101 not here?

111 → no.

100 from where?

101 → yes
111 → no
100 → no
000 → yes
110 → yes

(3,1) - repeating code

ENCODE

msg = 0 → code word: 000

msg = 1 → code word: 111

DECODE

111 → 1
010 → 0
011 → 1
000 → 0

?

Hamming distance

$d(u,v)$

$d(101, 100) = 1$ → $\underbrace{100 \quad 101}_{1\,?}$

? $\begin{cases} d(110, 001) = 3 \\ \downarrow \\ \underline{001}, 010, 011, 100, 101, 110 \end{cases}$

$d(101, 011) = 2$ (above: 3)

↓

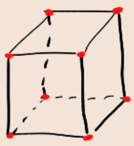
$011, \underbrace{100, 101}_{2}$

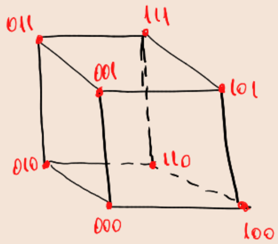
- $d(u,v) = d(v,u)$
- $d(u,v) + d(v,w) \geq d(u,w)$
- $d(u,v) \geq 0$

$d(u,v) = 0 \iff \underline{u = v}$

$(d(u,u) = 0)$

$(n-k)$ - code $\Rightarrow$ 2^n received words.





$d(001, 110)$ → $001 \to \underline{101 \to 111 \to 110}$
→ $001 \to \underline{011 \to 111 \to 110}$
→ $001 \to \underline{000 \to 100 \to 110}$
→ $001 \to \underline{000 \to 010 \to 110}$

~ Code $\underline{\text{detects}}$ t $(\leq t)$ errors $\iff$ $\boxed{\min(d(u,v)) \geq t+1}$

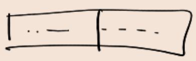
Code $\underline{\text{corrects}}$ - - - - - $\iff$ $\boxed{\min(d(u,v)) \geq 2t+1}$

$a_0 a_1 \dots a_{n-1}$ $\leadsto$ $\underline{a_0 + a_1 x + \dots + a_{n-1} \cdot x^{n-1}}$ $\in \mathbb{Z}_2[x]$ → $\{0,1\}$

p → generator of the code.

↓

generates polynomial code = (n,k) - code



↓

Code words: $\underline{\deg(\text{poly}) < n}$: p

$\xrightarrow{k}$ $\quad m \in \mathbb{Z}_2[x]$, $\deg(m) < k$.

① $p = 1 + x^2 + x^3 + x^4 \in \mathbb{Z}_2[x]$ generator poly of a $(7, 3)$ code. Encode $\underline{msg = 101}$.

$\underline{n = 7}$ $\quad$ $\underline{k = 3}$

msg $\underline{101}$ → $m = 1 \cdot 1 + 0 \cdot x + 1 \cdot x^2 = \underline{1 + x^2}$

→ $m \cdot x^{n-k} = (1+x^2) \cdot x^{7-3} = (1+x^2) \cdot x^4 = \underline{x^4 + x^6}$

→ $r = \underline{m \cdot x^{n-k}} \,\%\, p = (x^4 + x^6) \,\%\, (1 + x^2 + x^3 + x^4) = 1 + x$

$$
\begin{array}{l|l}
x^6 + x^4 & x^4 + x^3 + x^2 + 1 \\
\hline
-x^6 - x^5 - x^4 - x^2 & x^2 - x - 1 + 1 \\
\hline
-x^5 - x^2 & \\
x^5 + x^4 + x^3 + x &
\end{array}
$$

$$\begin{array}{l} /x^4+x^3-x^2+x \\ -x^4-x^3-x^2-1 \\ \hline / \ / -2x^2+x-1 \quad ? \end{array}$$

$\rightarrow r = 1+x+(x^4+x^6) = 1+x+x^4+x^6$

$\Rightarrow$ code word: $(1\,1\,0\,0|1\,0\,1)$

② $p = 1+x+x^3 \in \mathbb{Z}_2[x]$ gen. pol. of a $(6-3)$ code.

100011
100110 detectable errors?

100011 → code word? → poly : p?
100110

③ All the code words of $(6,3)$-code gen. by $p = 1+x+x^3 \in \mathbb{Z}_2[x]$.

$n=6$
$k=3$ $\rightarrow 2^k = 8$ code words.

000	→	000000
001	→	111001
010	→	011010
011	→	100011
100	→	010100
101	→	101101
110	→	001110
111	→	110111

how are they obtained?

$110 \Rightarrow m = 1+x$

$\rightarrow m \cdot x^{n-k} = (1+x)\cdot x^3 = x^3+x^4$

$\rightarrow r = m\cdot x^{n-k} \,\%\, p = (x^3+x^4) \,\%\, (1+x+x^3) = 1+x^2$

$\rightarrow r = r + m\cdot x^{n-k} = 1+x^2+x^3+x^4$

$$\begin{array}{r|l} x^4+x^3 & x^3+x+1 \\ \cline{2-2} -x^4-x^3-x & x \\ \hline / \ -x & \end{array}$$

??

Seminar 12

(3,2) – parity check code

2 digits message
3 digits code
first digit = (sum of the 2 digits in the message) %2

① a) 110 → $1 \overset{?}{=} (1+0)\%2 = 1\%2 = 1$ ⇒ no detectable errors.
010 → $0 \overset{?}{=} (1+0)\%2 = 1\%2 = 1$ ⇒ detectable error.
001 → $0 = (0+1)\%2 = 1\%2 = 1$ ⇒ " — " ——
111 → $1 = 2\%2 = 0$ ⇒ " — " ——
101 → $1 = 1\%2 = 1$ ⇒ no det. errors
000 → no det. errors.

Which digit is the message?

b) (3,1) – repeating code
1 digit message
3 digits code
first, second digits repeat the code.

111 → 11 repeat the message 1
011 → 01 repeat the message 1
101 → 10 repeat " — " —
010 → 01 repeat ?
000 → 00 repeat the message 0?
001 → 00 does not repeat the message 1

So the message is the last digit?

② $f = x^7+x^6+x^4+x^3+1$
$g = x^6+x^3+x^2+x$
code words in the poly (8,4)-code generated by $p = 1+x^2+x^3+x^4 \in \mathbb{Z}_2[x]$?

$n=8$
$k=4$

f : p

$$\begin{array}{l|l}
x^7+x^6+x^4+x^3+1 & x^4+x^3+x^2+1 \\
-x^7-x^6-x^5-x^3 & x^3+x \\
\hline
-x^5+x^4+1 & \\
x^5+x^4+x^3+x & \\
\hline
x^3+x+1 &
\end{array}$$

$x^3+x+1 \neq 0 \Rightarrow p \nmid f \Rightarrow f$ not a code word.

g : p

$$\begin{array}{l|l}
x^6+x^3+x^2+x & x^4+x^3+x^2+1 \\
-x^6-x^5-x^4-x^2 & x^2-x \\
\hline
-x^5-x^4+x^3+x & \\
x^5+x^4+x^3+x & \\
\hline
\underbrace{2x^3+2x}_{\mathbb{Z}_2!} = 0 &
\end{array}$$

⇒ g is a code word.

③ (6,3)-code $p = 1+x^2+x^3 \in \mathbb{Z}_2[x]$
$n=6$
$k=3$

000
001

$\boxed{m = 000} = 0\cdot x^0 + 0\cdot x^1 + 0\cdot x^2 = 0.$
$m \cdot x^{n-k} = 0.$
$r = m \cdot x^{n-k} \% p = 0$

$\underline{2^k} = 2^3 = 8$ words. $\left\{ \begin{array}{l} 010 \checkmark \\ 011 \checkmark \\ 100 \checkmark \\ 101 \\ 110 \\ 111 \end{array} \right.$

$r = m \cdot x^{n-k} \% p$

$v = r + m \cdot x^{n-k} = 0.$

$\Rightarrow \underbrace{000000}_{n \text{ digits}}$

$\boxed{001}$:

$m = 001 = 0 \cdot x^0 + 0 \cdot x^1 + 1 \cdot x^2 = x^2.$

$m \cdot x^{n-k} = x^2 \cdot x^{6-3} = x^2 \cdot x^3 = x^5.$

$r = m \cdot x^{n-k} \% p = x^5 \% (x^3 + x^2 + 1) = x + 1$, $v = r + mx^{n-k} = x + 1 + x^5 = \underline{1}\,\underline{1}\,\underline{0}\,\underline{0}\,\underline{0}\,\underline{1}$

$$
\begin{array}{l|l}
\quad x^5 & x^3 + x^2 + 1 \\ \cline{2-2}
-x^5 - x^4 - x^2 & x^2 - x + 1 \\ \cline{1-1}
-x^4 - x^2 & \\
\;\; x^4 + x^3 + x & \\ \cline{1-1}
\;\; x^3 - x^2 + x & \\
-x^3 - x^2 - 1 & \\ \cline{1-1}
\underbrace{-2x^2}_{0} + x - 1 = x + 1 &
\end{array}
$$

$-1 \equiv 1 \pmod 2$

$\boxed{010}$: $m = 010 = 0 \cdot x^0 + 1 \cdot x^1 + 0 \cdot x^2 = x.$

$m \cdot x^{n-k} = x \cdot x^3 = x^4.$

$r = m \cdot x^{n-k} \% p = x^4 \% (x^3 + x^2 + 1) = x^2 + x + 1.$

$$
\begin{array}{l|l}
\quad x^4 & x^3 + x^2 + 1 \\ \cline{2-2}
-x^4 - x^3 - x & x - 1 \\ \cline{1-1}
\;\; -x^3 - x & \\
\;\; x^3 + x^2 + 1 & \\ \cline{1-1}
\;\; x^2 - x + 1 = x^2 + x + 1 &
\end{array}
$$

$v = r + m \cdot x^{n-k} = x^4 + x^2 + x + 1.$

$\underline{1}\,\underline{1}\,\underline{1}\,\underline{0}\,\underline{1}\,\underline{0}$

$\boxed{011}$: $m = 011 = 0 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 = x + x^2.$

$m \cdot x^{n-k} = (x + x^2) \cdot x^3 = x^4 + x^5.$

?

$r = (m \cdot x^{n-k}) \% p = (x^5 + x^4) \% (x^3 + x^2 + 1) = x^2$

$$
\begin{array}{l|l}
\quad x^5 + x^4 & x^3 + x^2 + 1 \\ \cline{2-2}
-x^5 - x^4 - x^2 & x^2 \\ \cline{1-1}
\quad -x^2 = x^2 &
\end{array}
$$

$x^2 (x^3 + x^2 + 1) = x^5 + x^4 + x^2.$

$\boxed{100}$ $m = 1 \cdot x^0 + 0 \cdot x^1 + 0 \cdot x^2 = 1.$

$m \cdot x^{n-k} = 1 \cdot x^3 = x^3.$

$r = x^3 \% (x^3 + x^2 + 1) = x^2 + 1.$

$v = r + m \cdot x^{n-k} = x^3 + x^2 + 1.$

$\underline{1}\,\underline{0}\,\underline{1}\,\underline{1}\,\underline{0}\,\underline{0}$

$$
\begin{array}{l|l}
\quad x^3 & x^3 + x^2 + 1 \\ \cline{2-2}
-x^3 - x^2 - 1 & 1 \\ \cline{1-1}
\;\; -x^2 - 1 & \\
\;\; = x^2 + 1 &
\end{array}
$$

$\boxed{101}$ $m = 1 \cdot x^0 + 0 \cdot x^1 + 1 \cdot x^2 = x^2 + 1.$

$m \cdot x^3 = (x^2 + 1) x^3 = x^5 + x^3.$

$r = (x^5 + x^3) \% (x^3 + x^2 + 1) = x^2 + x$ $\quad v = x^5 + x^3 + x^2 + x$

$$
\begin{array}{l|l}
x^5 + x^3 & x^3 + x^2 + 1 \\
\hline
-x^5 - x^4 - x^2 & x^2 - x + 2 \quad = \underline{x^2 + x} \\
\hline
-x^4 + x^3 - x^2 & \\
\quad x^4 + x^3 + x & \\
\hline
2x^3 - x^2 + x & \\
-2x^3 - 2x^2 - 2 & \\
\hline
-3x^2 + x - 2 \quad = -x^2 + x = \underline{x^2 + x}. &
\end{array}
$$

0 1 1 1 0 1

(4) $G = \left(\frac{P}{I_3}\right) \in \mathcal{M}_{5,3}(\mathbb{Z}_2)$ | Parity Check matrix? Code words?

$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

$\begin{matrix} n=5 \\ k=3 \end{matrix} \Rightarrow n-k = 5-3 = 2$

$H = (I_{n-k} \mid P) = (I_2 \mid P) = \left(\begin{array}{cc|ccc} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{array}\right)$

$u = (u_1, u_2, u_3, u_4, u_5)$ code vector if $\underline{H \cdot u = 0_2}$

$\Rightarrow \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix}}_{2\times 5} \cdot \underbrace{(u_1, u_2, u_3, u_4, u_5)}_{5\times 1} = \underbrace{\begin{pmatrix} 0 \\ 0 \end{pmatrix}}_{2\times 1}$

$\Rightarrow \begin{pmatrix} u_1 & 0 & 0 & 0 & u_5 \\ 0 & u_2 & u_3 & u_4 & u_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\Rightarrow \begin{cases} u_1 + u_5 = 0 & \Rightarrow \underline{u_5 = -u_1}. \\ u_2 + u_3 + u_4 + u_5 = 0. & \Rightarrow u_2 + u_3 + u_4 - u_1 = 0. \\ & \Rightarrow \underline{u_1 = u_2 + u_3 + u_4} \end{cases}$

$\Rightarrow u = (u_1, u_2, u_3, u_4, u_5) = (u_2 + u_3 + u_4, u_2, u_3, u_4, \underline{-u_2 - u_3 - u_4}).$

?

$\Rightarrow 2^{k=3}$ code words. (8).

$\Rightarrow \{(0,0,0,0,0), (1, 1,0,0, \underset{}{\overset{= 1}{-1}}), (1, 0,1,0, \overset{= 1}{-1}), (0,0,0,1,1), (\underset{\downarrow\, 0}{2}, 1,1,0,0),$

$(\underset{= 0}{2}, 0,1,1,0), (\underset{= 0}{2}, 1,0,1,0), (\underset{= 1}{3}, 1,1,1, \underset{= 1}{-1})\}$

Recall.

(3,2)-party check code
- 2 digits message
- 3 digits code (encode)
- 1st digit = sum of the 2 digits in the message

(3,1)-repeating code
- 1 digit message
- 3 digits code
- 1st, 2nd digits repeat the code

Party-check matrix

$H = (I_{n-k} \mid P)$

$u \in \mathcal{M}_{n,1}(\mathbb{Z}_2)$ is a code vector

$\Leftrightarrow H \cdot u = 0.$

(mod 2)

Generator of a polynomial code

$p \in \mathbb{Z}_2[x]$ of degree $n-k$

$\downarrow$

generator of a polynomial code (n,k), whose words are polynomials of degree less than n, $\vdots\, p$.

$\downarrow$

for a $(n-k)$ polynomial code $\to 2^k$ code words.

for a message m : $\boxed{m \cdot x^{n-k} = q p + r}$ degree$(r) <$ degree$(p) = n-k$.

$\hookrightarrow$ we code it as : $v = r + m \cdot x^{n-k}$.

Hamming distance

u, v of the same length $\Rightarrow$ the nr. of positions in which they differ. $\to d(u,v) \to$ metric on $\mathbb{Z}_2^n$.

A code detects all errors $\leq t$:

$\boxed{\min(d(u,v)) \geq t+1}$.

A code corrects all errors $\leq t$:

$\boxed{\min(d(u,v)) \geq 2t+1}$

Encoder

$\gamma : \mathbb{Z}_2^k \to \mathbb{Z}_2^n$ with $[\gamma]_{EE'} = G$.

5. $G = \left(\frac{P}{I_4}\right) \in \mathcal{M}_{9,4}(\mathbb{Z}_2)$. $\to 2^4 = 16$ code words?

$$H = (I_5 | P) \Rightarrow H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

For $u = (u_1, \dots, u_9)$ we compute $H \cdot u = 0_9$

$$\Rightarrow \begin{cases} u_1 + u_8 = 0 \Rightarrow u_1 = -u_8 \Rightarrow \underline{u_1 = u_8}. \\ u_2 + u_7 + u_9 = 0 \Rightarrow u_2 = -u_7 - u_9 = \underline{u_7 + u_9}. \\ u_3 + u_6 + u_8 + u_9 = 0 \Rightarrow u_3 = -u_6 - u_8 - u_9 = \underline{u_6 + u_8 + u_9}. \\ u_4 + u_7 = 0 \Rightarrow u_4 = -u_7 = \underline{u_7}. \\ u_5 + u_6 + u_9 = 0 \Rightarrow u_5 = -u_6 - u_9 = u_6 + u_9. \end{cases}$$

$\Rightarrow u = (u_8, u_7 + u_9, u_6 + u_8 + u_9, u_7, \underline{u_6 + u_9}, u_6, u_7, u_8, u_9)$

$\min(d(u_i, u_j))$

$\min(d(u_1, u_i)) = \min(d(u_2, u_i)) = \dots = \min(d(u_9, u_i)) = 3$.

$\min(d(u_i, u_j)) = 3 \geq t+1$

$\Rightarrow t \leq 2 \Rightarrow$ code detects 2 errors.

$\min(d(u_i, u_j)) = 3 \geq 2t+1$

$\Rightarrow t \leq 1 \Rightarrow$ code corrects 1 error.

6. Encode $\begin{cases} 1101 \\ 0111 \\ 0000 \end{cases}$ using the generator matrix of the $(9,4)$-code of Ex 5.

$\begin{cases} 0000 \\ 1000 \end{cases}$

$\boxed{[\gamma]_{EE'} = G}$ $\Rightarrow$ $\begin{cases} \gamma(e_1) = \gamma(1000) \dots & 001011000 \\ \gamma(e_2) = \gamma(0100) \dots & 010100100 \\ \gamma(e_3) = \gamma(0010) \dots & 101000010 \\ \gamma(e_4) = \gamma(0001) \dots & 011010001 \end{cases}$

$1101 = e_1 + e_2 + e_4 \Rightarrow \gamma(1101) = \gamma(e_1+e_2+e_4) \overset{\text{WHY}}{=\!=\!=} \gamma(e_1) + \gamma(e_2) + \gamma(e_4)$

$= 001011000 + 010100100 + 011010001$

$= 000101101.$

$0111 = e_2 + e_3 + e_4 \Rightarrow \gamma(0111) = \gamma(e_2) + \gamma(e_3) + \gamma(e_4) = 100110111.$

$0000 = e_1 + e_2 \quad \Rightarrow \dots$

$1000 = e_1 \quad \Rightarrow \dots$

(7) Generator matrix =? Parity check matrix?

$(4, 1)$ – code gen. by $p = 1 + x + x^2 + x^3 \in \mathbb{Z}_2[x]$.

$\gamma : \mathbb{Z}_2^{1} \to \mathbb{Z}_2^{4}$

$\begin{cases} \text{every codeword has } n \text{ bits } (4). \\ k = 1 \to \text{choose 1 message bit } (k) \end{cases}$

$[\gamma]_{EE'} = G$

a 4-bit word (c_0, c_1, c_2, c_3)

$E = (e_1) = 1$

$E' = (e_1', e_2', e_3', e_4')$

$c_0 + c_1 x + c_2 x^2 + c_3 x^3.$

$p = 1 + x + x^2 + x^3$ corresponds to 1111 = message

$e_1 = 1 \Rightarrow m = 1 \Rightarrow m \cdot x^{n-k} = 1 \cdot x^3 = x^3$

$r = (m \cdot x^{n-k}) \,\%\, p = x^3 \,\%\, (1 + x + x^2 + x^3) = x^2 + x + 1.$

$$\begin{array}{l|l} x^3 & x^3 + x^2 + x + 1 \\ \hline -x^3 - x^2 - x - 1 & 1 \\ \hline -x^2 - x - 1 & \\ \quad \| & \\ x^2 + x + 1 & \end{array}$$

$v = x^3 + x^2 + x + 1 \Rightarrow \underline{1111}.$

$\Rightarrow G = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} P \\ I_k \end{bmatrix} \Rightarrow P = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$H = \underset{I_3}{(I_{n-k} \mid P)} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

(8) Generator matrix =?
Parity check matrix?

$(7, 3)$ – code generated by $p = 1 + x^2 + x^3 + x^4 \in \mathbb{Z}_2[x]$.

$\gamma: \mathbb{Z}_2 \to \mathbb{Z}_2$

$[\gamma]_{EE'} = G$

$E = (e_1, e_2, e_3)$

$E' = (e_1', e_2', e_3', e_4', e_5', e_6', e_7')$

$e_1 = (1,0,0) \Rightarrow 100 = 1\cdot x^0 + 0\cdot x^1 + 0 \cdot x^2 = 1 = m$

$m \cdot x^{n-k} = 1 \cdot x^4 = x^4.$

$r = x^4 \,\%\, (x^4+x^3+x^2+1) = x^3+x^2+1.$

$$
\begin{array}{r|l}
x^4 & x^4+x^3+x^2+1 \\ \cline{2-2}
-x^4-x^3-x^2-1 & 1 \\ \cline{1-1}
-x^3-x^2-1 & \\
\shortparallel & \\
x^3+x^2+1. &
\end{array}
$$

$v = x^4+x^3+x^2+1.$

$\Rightarrow 1\ 0\ 1\ 1\ 1\ 0\ 0$

$e_2 = (0,1,0) \Rightarrow 010 = 0\cdot x^0 + 1 \cdot x^1 + 0\cdot x^2 = x.$

$m \cdot x^{n-k} = x \cdot x^4 = x^5.$

$r = x^5 \,\%\, (x^4+x^3+x^2+1) = x^2+x+1$

$$
\begin{array}{r|l}
x^5 & x^4+x^3+x^2+1 \\ \cline{2-2}
-x^5-x^4-x^3-x & x-1 \\ \cline{1-1}
-x^4-x^3-x & \\
x^4+x^3+x^2+1 & \\ \cline{1-1}
x^2-x+1 & \\
x^2+x+1 &
\end{array}
$$

$v = 1+x+x^2+x^5$

$1\ 1\ 1\ 0\ 0\ 1\ 0$

$e_3 = (0,0,1) \Rightarrow 001 = 0\cdot x^0 + 0\cdot x^1 + 1\cdot x^2 = x^2 = m$

$m \cdot x^{n-k} = x^2 \cdot x^4 = x^6$

$r = x^6 \,\%\, (x^4+x^3+x^2+1) = x^3+x^2+x.$

$$
\begin{array}{r|l}
x^6 & x^4+x^3+x^2+1 \\ \cline{2-2}
-x^6-x^5-x^4-x^2 & x^2-x \\ \cline{1-1}
-x^5-x^4-x^2 & \\
x^5+x^4+x^3+x & \\ \cline{1-1}
/\ x^3-x^2+x & \\ \cline{1-1}
=x^3+x^2+x. &
\end{array}
$$

$v = x^6+x^3+x^2+x.$

$0\ 1\ 1\ 1\ 0\ 0\ 1$

$$
G = \begin{bmatrix}
1 & 1 & 0 \\
0 & 1 & 1 \\
1 & 1 & 1 \\
1 & 0 & 1 \\ \hdashline
1 & 0 & 0 \\
0 & 1 & 0 \\
0 & 0 & 1
\end{bmatrix}
\begin{array}{l}
\\ P \text{ (parity check matrix)} \\ \\ \\ \\ I_3\ (I_k) \\ \\
\end{array}
$$

generator matrix

$$\Rightarrow P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

parity check matrix

$$\Rightarrow H = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$\uparrow$ I_{n-k} $\uparrow$ I_4 $\qquad$ $\uparrow$ P